

Project Initialization and Planning Phase

Date	30 September 2026
Team ID	SWTID-2026-5807
Project Title	AI-StudyBuddy - Personalized AI Study Assistant for Students
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform the way students study using artificial intelligence, improving understanding and exam readiness. It tackles scattered study material and the lack of personal guidance, promising saved time, reduced stress and better results. Key features include an AI doubt-solving chatbot, notes summarizer, quiz generator and smart study planner.

Project Overview	
Objective	To build an AI-powered study assistant that helps students clear doubts instantly, revise faster and stay organized through personalized summaries, quizzes and study plans.
Scope	The project covers a web application with an AI doubt-solving chatbot, notes summarizer, quiz and flashcard generator, study planner with reminders, and a progress dashboard.
Problem Statement	
Description	Students struggle to clear doubts instantly, revise from scattered and lengthy material, and follow a consistent study plan, leading to stress and poor exam performance.
Impact	Solving this will save study time, improve understanding and exam confidence, and give every student affordable, personalized guidance anytime, anywhere.
Proposed Solution	
Approach	Using a large language model and NLP, AI-StudyBuddy understands student queries, summarizes content, generates practice questions and adapts study plans to each student's pace and weak topics.
Key Features	- AI chatbot for instant, 24/7 doubt solving and explanations. - Notes / PDF summarizer for quick revision.

	- Auto-generated quizzes and flashcards on any topic. - Personalized study planner with smart reminders. - Progress dashboard that highlights weak topics.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	2 vCPU cloud server (AI runs via LLM API, so no local GPU needed)
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	50 GB SSD (cloud storage)
Software		
Frameworks	Python frameworks	React.js, FastAPI
Libraries	Additional libraries	PyPDF, pandas, numpy, scikit-learn, LangChain
Development Environment	IDE	VS Code, Postman, Jupyter Notebook
Data		
Data	Source, size, format	User-uploaded notes (PDF/TXT), open Q&A datasets (JSON)